

# Per-SYT vanishing at the leftmost factor: $p_T(q) = 0$ whenever pair $(\ell, \ell + 1)$ is same-column in $T$

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## Abstract

For  $\lambda = (\widehat{\lambda}, 1^q) \vdash n$  with  $\ell = \ell(\lambda) = \ell(\widehat{\lambda}) + q$ , let  $\Omega^{(\lambda)} = R'_{\ell+1} R'_{\ell+2} \cdots R'_n$  be the column-1 chain product in the Iwahori–Hecke algebra  $H_q(S_n)$ , and let  $p_T(q) := \langle v_T, \Omega v_T \rangle$  be the diagonal matrix coefficient of  $\Omega$  in the Hoefsmit ( $b' = 1$ ) seminormal basis. We prove the following per-SYT vanishing result: *if the values  $\ell$  and  $\ell + 1$  sit in the same column of  $T$  (equivalently,  $S_\ell v_T = 0$ ), then  $p_T(q) = 0$ .* The proof is two lines once the leftmost-factor identity  $\Omega = S_\ell \cdot X$  is in place:  $\Omega v_T \in \text{im}(S_\ell) = E_\ell^+$  while  $v_T \in E_\ell^-$ , and the seminormal-basis projection onto  $E_\ell^-$  extracts the diagonal coefficient cleanly. The result is a per-SYT refinement of the universal eigenspace statement  $B^{(\lambda)} = \Omega V^\lambda \subseteq E_\ell^+$  obtained from the same leftmost-factor identity (May-13 evening-2, prior memo). We verify the theorem on seven shapes spanning  $\tau = 0$  and  $\tau \geq 1$  and use it to account for 26 out of 108 zero diagonals at the smoking-gun shape  $\lambda = (3, 3, 1, 1, 1)$ , including the historical mystery tableau  $T_0 = ((1, 2, 5), (3, 4, 6), (7), (8), (9))$ . The remaining 82 zeros at this shape arise from a finer mechanism left open for future work.

## 1 Setup

We adopt the conventions of the trace-vanishing papers [1, 2, 3].

$H_q(S_n)$  is the Iwahori–Hecke algebra over  $\mathbb{Q}(q)$  with generators  $T_1, \dots, T_{n-1}$ , braid relations  $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$ , commutation  $T_i T_j = T_j T_i$  for  $|i - j| \geq 2$ , and quadratic  $T_i^2 = (q - 1)T_i + q$ . We set  $S_i := T_i + 1$ , so the Hecke quadratic is equivalent to

$$S_i^2 = (q + 1)S_i, \quad T_i S_i = q S_i = S_i T_i. \quad (1)$$

The image  $\text{im}(S_i)$  is the  $+q$ -eigenspace  $E_i^+ := \ker(T_i - q)$ , and the kernel  $\ker(S_i)$  is the  $-1$ -eigenspace  $E_i^- := \ker(T_i + 1)$ . Both are well-defined on any  $H_q$ -module, and on the (finite-dimensional) Specht module  $V^\lambda$  one has the direct-sum decomposition

$$V^\lambda = E_i^+|_{V^\lambda} \oplus E_i^-|_{V^\lambda} \quad (2)$$

since  $S_i$  is diagonalizable on  $V^\lambda$  with eigenvalues  $\{q + 1, 0\}$  (this follows immediately from (1): every  $x \in V^\lambda$  decomposes as  $\frac{1}{q+1} S_i x + (x - \frac{1}{q+1} S_i x)$  with the first summand in  $E_i^+$  and the second in  $E_i^-$ ).

For a SYT  $T$  of shape  $\lambda$ ,  $v_T$  denotes the Hoefsmit seminormal basis vector, normalised with  $b' = 1$  as in [2]. The local action of  $S_i$  on  $v_T$  depends only on the relative positions of  $i$  and  $i + 1$  in  $T$ :

- **Same row (SR):**  $S_i v_T = (q + 1)v_T$ , so  $v_T \in E_i^+$ ;
- **Same column (SC):**  $S_i v_T = 0$ , so  $v_T \in E_i^-$ ;

- **Diagonal** (D, neither):  $S_i$  acts on the 2-dim block  $\{v_T, v_{s_i T}\}$  as a non-degenerate matrix; both  $v_T$  and  $v_{s_i T}$  have nonzero components in both  $E_i^+$  and  $E_i^-$ , but the  $E_i^-$  component is contained in the same block  $\text{span}(v_T, v_{s_i T})$ .

For  $k \geq 1$ ,  $R'_k := S_{k-1} S_{k-2} \cdots S_1$ , with  $R'_1 = 1$ . The chain product on  $V_\lambda$  is

$$\Omega^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n,$$

where  $\ell = \ell(\lambda)$ . We write  $B^{(\lambda)} = \Omega V^\lambda$  for the chain image and  $p_T(q) := \langle v_T, \Omega v_T \rangle$  for the  $(T, T)$ -diagonal coefficient. We use  $\langle v_S, v_T \rangle = \delta_{S,T}$  as the bilinear pairing implicit in “matrix coefficient” (equivalently, the coefficient of  $v_T$  in the seminormal expansion of  $\Omega v_T$ ).

## 2 Statement

**Theorem 1** (Per-SYT leftmost-factor sign-kill). *Let  $\lambda \vdash n$  with  $\ell = \ell(\lambda)$ , and let  $\Omega = R'_{\ell+1} R'_{\ell+2} \cdots R'_n$  on  $V^\lambda$ . For any  $T \in \text{SYT}(\lambda)$ , if the values  $\ell$  and  $\ell + 1$  sit in the same column of  $T$  (equivalently, the pair  $(\ell, \ell + 1)$  is SC at  $T$ , i.e.,  $S_\ell v_T = 0$ ), then*

$$p_T(q) = \langle v_T, \Omega v_T \rangle = 0 \quad \text{in } \mathbb{Q}(q).$$

The hypothesis is purely a statement about  $T$  (a placement of the values  $\ell, \ell + 1$ ); the conclusion is about the chain operator. The theorem does not require any restriction on  $\lambda$  (such as multi-row,  $\tau = 0$ , or staircase shape). It is the per-SYT refinement of the universal containment  $B^{(\lambda)} \subseteq E_\ell^+|_{V^\lambda}$ .

## 3 Proof

### The leftmost-factor identity

**Lemma 2** (Leftmost factor).  $\Omega = S_\ell \cdot X$  for some  $X \in H_q(S_n)$ . Consequently,  $\Omega v \in \text{im}(S_\ell) = E_\ell^+|_{V^\lambda}$  for every  $v \in V^\lambda$ .

*Proof.*  $R'_{\ell+1} = S_\ell S_{\ell-1} \cdots S_1 = S_\ell \cdot R'_\ell$ . So

$$\Omega = R'_{\ell+1} R'_{\ell+2} \cdots R'_n = S_\ell \cdot (R'_\ell R'_{\ell+2} \cdots R'_n) = S_\ell \cdot X$$

with  $X = R'_\ell R'_{\ell+2} \cdots R'_n$ . Then  $\Omega v = S_\ell(Xv) \in \text{im}(S_\ell)$ . Since  $\text{im}(S_\ell) \subseteq E_\ell^+$  on every module (by  $T_\ell S_\ell = q S_\ell$ , so any  $S_\ell w$  satisfies  $T_\ell(S_\ell w) = q(S_\ell w)$ ), the conclusion follows.  $\square$   $\square$

### The projection argument

*Proof of Theorem 1.* Suppose pair  $(\ell, \ell + 1)$  is SC at  $T$ , so  $v_T \in E_\ell^-|_{V^\lambda}$ . By Lemma 2,  $\Omega v_T \in E_\ell^+|_{V^\lambda}$ .

By (2),  $V_\lambda = E_\ell^+ \oplus E_\ell^-$ . Let  $\pi_- : V_\lambda \rightarrow E_\ell^-$  be the projection along  $E_\ell^+$ . Then  $\pi_-(\Omega v_T) = 0$ .

Expand  $\Omega v_T$  in the seminormal basis:

$$\Omega v_T = \sum_{T' \in \text{SYT}(\lambda)} c_{T,T'} v_{T'}, \tag{3}$$

with  $c_{T,T'} \in \mathbb{Q}(q)$  and  $c_{T,T} = p_T(q)$ . Apply  $\pi_-$ :

$$0 = \pi_-(\Omega v_T) = \sum_{T'} c_{T,T'} \pi_-(v_{T'}). \tag{4}$$

Partition  $\text{SYT}(\lambda)$  by the type of pair  $(\ell, \ell + 1)$ :

- $T' \in \text{SYT}^{\text{SR}}$ :  $v_{T'} \in E_\ell^+$ , so  $\pi_-(v_{T'}) = 0$ .
- $T' \in \text{SYT}^{\text{SC}}$ :  $v_{T'} \in E_\ell^-$ , so  $\pi_-(v_{T'}) = v_{T'}$ .
- $T' \in \text{SYT}^{\text{D}}$ :  $v_{T'}$  lies in the 2-dim block  $W_{T'} := \text{span}(v_{T'}, v_{s_\ell T'})$ , with  $s_\ell T' \in \text{SYT}^{\text{D}}$  also. The block decomposes as  $W_{T'} = (W_{T'} \cap E_\ell^+) \oplus (W_{T'} \cap E_\ell^-)$ , each summand 1-dimensional. There exist  $\alpha_{T'}, \beta_{T'} \in \mathbb{Q}(q)$  (depending only on the local  $T_\ell$ -axial distance at  $T'$ ) with  $\pi_-(v_{T'}) = \alpha_{T'} v_{T'} + \beta_{T'} v_{s_\ell T'}$ .

Substituting into (4):

$$0 = \underbrace{\sum_{T' \in \text{SYT}^{\text{SC}}} c_{T,T'} v_{T'}}_{(\star)} + \sum_{T' \in \text{SYT}^{\text{D}}} c_{T,T'} (\alpha_{T'} v_{T'} + \beta_{T'} v_{s_\ell T'}). \quad (5)$$

The seminormal basis  $\{v_{T'}\}_{T' \in \text{SYT}(\lambda)}$  is linearly independent. Read off the coefficient of  $v_{T''}$  in (5) for each  $T'' \in \text{SYT}(\lambda)$ :

- If  $T'' \in \text{SYT}^{\text{SR}}$ :  $v_{T''}$  does not appear on the right (it is not in any  $E_\ell^-$  block). Coefficient: 0. No constraint on the  $c_{T,\cdot}$ .
- If  $T'' \in \text{SYT}^{\text{SC}}$ :  $v_{T''}$  appears only as the  $T'' = T'$  term of the  $(\star)$  sum (it is not in any D-block since  $s_\ell T''$  is not an SYT). Coefficient:  $c_{T,T''}$ .
- If  $T'' \in \text{SYT}^{\text{D}}$ :  $v_{T''}$  appears as the self-term of  $T' = T''$  in the D-sum (contributing  $c_{T,T''} \alpha_{T''}$ ) and as the partner-term of  $T' = s_\ell T''$  (contributing  $c_{T,s_\ell T''} \beta_{s_\ell T''}$ ). Coefficient:  $c_{T,T''} \alpha_{T''} + c_{T,s_\ell T''} \beta_{s_\ell T''}$ .

Setting each coefficient to zero:

$$c_{T,T''} = 0 \quad \text{for every } T'' \in \text{SYT}^{\text{SC}}, \quad (6)$$

and one linear relation per D-block on  $(c_{T,T''}, c_{T,s_\ell T''})$ .

By hypothesis  $T \in \text{SYT}^{\text{SC}}$ . Substituting  $T'' = T$  in (6):

$$p_T(q) = c_{T,T} = 0. \quad \square$$

$\square$

## 4 Remarks

### 4.1 Why the leftmost factor is structurally privileged

Lemma 2 extracts  $S_\ell$  as the *leftmost* factor of  $\Omega$ . The same trick cannot be applied to inner factors  $S_j$  with  $j > \ell$ , because  $S_j$  is sandwiched between sub-words of  $\Omega$  that contain  $S_{j-1}$  and  $S_{j+1}$ , and braid/commutation relations do not permit moving  $S_j$  to the leftmost position without picking up correction terms. The symmetric “rightmost factor” identity  $\Omega = Y \cdot S_1$  — where  $Y = R'_{\ell+1} \cdots R'_{n-1} \cdot S_{n-1} S_{n-2} \cdots S_2$  — gives *strict* vanishing  $\Omega v_T = 0$  when pair  $(1, 2)$  is SC at  $T$  (descent-1): then  $S_1 v_T = 0$ , so  $\Omega v_T = Y \cdot 0 = 0$ . This is a trivial consequence and is independent of Theorem 1; the leftmost analysis is genuinely deeper because it gives  $p_T = 0$  from *coefficient* information (not from  $\Omega v_T = 0$ , which need not hold under the SC-at- $(\ell, \ell + 1)$  hypothesis).

## 4.2 Combination with Pillar 1

The Pillar 1 meta-theorem [3] establishes  $B^{(\lambda)} \subseteq E_j^-|_{V^\lambda}$  for  $j = 1, 2, \dots, \tau(\widehat{\lambda})$  under the standing hypothesis. Run the same projection argument at each such  $j$ : if  $v_T \in E_j^+$  (i.e., pair  $(j, j+1)$  is SR at  $T$ ), then  $c_{T, T''}$  vanishes for every  $T'' \in \text{SYT}^{\text{SR}} \text{ at } j$ . In particular, taking  $T'' = T$  yields:

**Corollary 3** (Pillar-1-side per-SYT vanishing). *For  $\lambda = (\widehat{\lambda}, 1^q)$  under the standing hypothesis of Pillar 1 ( $\ell(\widehat{\lambda}) \geq 2$ , every row  $\widehat{\lambda}_r \geq 2$ ,  $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ ) and  $\tau := \tau(\widehat{\lambda})$ : if there exists  $j \in \{1, \dots, \tau\}$  with pair  $(j, j+1)$  same-row at  $T$ , then  $p_T(q) = 0$ .*

Combined, the two sides yield:

**Corollary 4** (Necessary support of  $p_T \neq 0$ ). *At  $\lambda = (\widehat{\lambda}, 1^q)$  in the Pillar-1 regime,  $p_T(q) \neq 0$  implies*

- pair  $(j, j+1)$  is not SR at  $T$  for any  $j = 1, \dots, \tau$  (Corollary 3), and
- pair  $(\ell, \ell+1)$  is not SC at  $T$  (Theorem 1).

For descent-0 shapes (i.e.,  $T$  with pair  $(1, 2)$  SR), the trace  $\text{tr}_{V^\lambda}(\Omega) = \sum_T p_T(q)$  is concentrated on  $T$ 's satisfying the second bullet point. At  $\tau = 0$ , this is the only “free” constraint and Theorem 1 is the main per-SYT vanishing mechanism known.

## 5 Computational verification

We verify Theorem 1 by enumerating  $\text{SYT}(\lambda)$  and computing  $p_T(q)$  symbolically in the Hoefsmit ( $b'=1$ ) basis, using the framework from [2]. For each shape, we count  $N_{\text{SC}}(\ell) := |\{T : \text{pair } (\ell, \ell+1) \text{ SC at } T\}|$ ,  $N_0^{\text{SC}}(\ell)$  the count of those with  $p_T \equiv 0$ , and the total count  $N_0$  of zero diagonals across all  $T \in \text{SYT}(\lambda)$ . The theorem predicts  $N_{\text{SC}}(\ell) = N_0^{\text{SC}}(\ell)$ ; the “gap”  $N_0 - N_0^{\text{SC}}(\ell)$  measures vanishings not explained by Theorem 1.

Table 1: Verification of Theorem 1 on seven shapes spanning  $\tau = 0$  and  $\tau \geq 1$ . The  $N_{\text{SC}}(\ell) = N_0^{\text{SC}}(\ell)$  equality holds in every case (consistent with the theorem); no counterexamples were found. The “ $N_0$  (total)” column counts *all* zero diagonals; the gap  $N_0 - N_0^{\text{SC}}(\ell)$  measures vanishings arising from mechanisms beyond Theorem 1 (descent-1 trivial vanishing,  $\tau$ -side Pillar 1 sign-kill, and the open “interior cancellation” phenomenon).

| $\lambda$       | $n$ | $\ell$ | $ \text{SYT}(\lambda) $ | $N_{\text{SC}}(\ell)$ | $N_0^{\text{SC}}(\ell)$ | $N_0$ (total) |
|-----------------|-----|--------|-------------------------|-----------------------|-------------------------|---------------|
| (2, 2, 1)       | 5   | 3      | 5                       | 2                     | 2                       | 4             |
| (3, 2, 1, 1)    | 7   | 4      | 35                      | 8                     | 8                       | 30            |
| (4, 2, 1)       | 7   | 3      | 35                      | 6                     | 6                       | 18            |
| (2, 2, 2, 1, 1) | 8   | 5      | 28                      | 12                    | 12                      | 28            |
| (3, 3, 1, 1, 1) | 9   | 5      | 120                     | 26                    | 26                      | 108           |
| (3, 2, 2, 1, 1) | 9   | 5      | 162                     | 45                    | 45                      | 150           |
| (4, 3, 1, 1)    | 9   | 4      | 216                     | 30                    | 30                      | 147           |

Shapes (2, 2, 1) through (3, 2, 2, 1, 1) and (2, 2, 2, 1, 1) were verified symbolically over  $\mathbb{Q}(q)$ . Shape (4, 3, 1, 1) was verified numerically at  $q = 11$  and  $q = 13$  over  $\mathbb{Q}$  — two generic non-pole values; a nonzero  $p_T \in \mathbb{Q}(q)$  has finitely many roots, so vanishing at both these values is equivalent to symbolic vanishing modulo the (finite, known) seminormal denominator. Scripts: `scratch/2026-05-17-leftmost-fac`

**Smoking gun.** For  $\lambda = (3, 3, 1, 1, 1)$ , the historically mysterious zero diagonal at  $T_0 = ((1, 2, 5), (3, 4, 6), (7), (8), (9))$  is now *explained*: pair  $(5, 6)$  has 5 at  $(0, 2)$  and 6 at  $(1, 2)$  — same column — so Theorem 1 fires with  $\ell = 5$ , giving  $p_T[T_0] = 0$ .

## 6 Discussion

### 6.1 What this theorem explains

At  $\lambda = (3, 3, 1, 1, 1)$ , the agent sweep of 17 May 2026 found 108 zero diagonals out of 120 SYTs. Theorem 1 accounts for 26 — the ones where pair  $(5, 6)$  is same-column — and gives a structural reason:  $v_T \in E_5^-$  while  $\Omega v_T \in E_5^+$ , so the seminormal coefficient must vanish.

The remaining  $108 - 26 = 82$  zeros split further. A subset are *descent-1* SYTs (pair  $(1, 2)$  SC), for which  $\Omega v_T = 0$  already (the rightmost-factor argument of Section 4); these are  $f^{(2, 2, 1, 1, 1)} = 14$  tableaux. The remaining  $82 - 14 = 68$  are descent-0 SYTs with pair  $(5, 6)$  D — for example

$$T = ((1, 2, 3), (4, 5, 8), (6), (7), (9))$$

where pair  $(5, 6)$  is D (5 at  $(1, 1)$ , 6 at  $(2, 0)$ ). Theorem 1 does not fire for these — yet symbolic computation returns  $p_T = 0$ . The mechanism here is a higher-order cancellation in the seminormal expansion of  $\Omega v_T$ , possibly the “interior sign-kill” pattern noted in the May-16 dream-2 entry. Characterising these is an open question.

### 6.2 Relation to other techniques

Theorem 1 is to the per-SYT diagonal what the leftmost-factor memo [4, **leftmost-rightmost-factor-technique**] is to the chain image: the image  $B^{(\lambda)} = \Omega V^\lambda$  lies in  $E_\ell^+$ , and the per-SYT specialization extracts the diagonal coefficient from that universal containment. The proof uses only:

- (i) the Hecke quadratic in the form  $S_\ell^2 = (q + 1)S_\ell$ , which gives diagonalizability (2);
- (ii) the seminormal-basis local action of  $S_\ell$  on  $v_T, v_{s_\ell T}$ .

No external machinery (Jucys-Murphy, Markov trace, KL/canonical, BGG, 1423-avoidance, Goertzen-Williamson, etc.) is invoked. This places Theorem 1 squarely in the family of one-line algebraic facts that arise from Hecke absorption — and indeed extends the May-13 leftmost-rightmost-factor toolkit by one notch (per-SYT, not just  $B$ ).

### 6.3 What remains open

- **Characterising the 82 unexplained zeros at  $(3, 3, 1, 1, 1)$ .** These have pair  $(\ell, \ell + 1)$  D, so  $v_T$  has nonzero  $E_\ell^-$  and  $E_\ell^+$  components, and the projection-onto- $E_\ell^-$  argument no longer forces  $c_{T, T} = 0$  directly. A finer mechanism is needed.
- **Iterated leftmost factors.** Can the argument be iterated on  $X = R'_\ell R'_{\ell+2} \cdots R'_n$ , exposing a second-leftmost  $S_{\ell-1}$ ? The image of  $S_\ell S_{\ell-1}$  is a proper subspace of  $E_\ell^+$ , but its description is not transparent. A clean characterisation would mechanically extend Theorem 1.
- **D-block constraints.** The D-block linear relations in the proof — one per unordered pair  $\{T', s_\ell T'\}$  — are nontrivial constraints that the present theorem does *not* extract. They relate  $c_{T, T''}$  and  $c_{T, s_\ell T''}$  for D tableaux  $T''$ . A careful analysis of these constraints might yield further vanishing in the D regime, possibly explaining a fraction of the 82.

## References

- [1] Clio, *Trace-vanishing forward direction: corollary of multi-row meta + right-absorption*, 16 May 2026. `~/projects/proofs/2026-05-16-trace-vanishing-forward.tex`.
- [2] Clio, *Converse trace-vanishing: per-SYT positivity and column-descent-free SYT existence*, 16 May 2026. `~/projects/proofs/2026-05-17-converse-per-syt-positivity.tex`.
- [3] Clio, *Extended sign-kill for general multi-row  $\hat{\lambda}$ : the meta-theorem at arbitrary  $\ell(\hat{\lambda}) \geq 2$* , 15 May 2026. `~/projects/proofs/2026-05-15-multi-row-sign-kill-meta.tex`.
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